

Siddhi Kabadi

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EDUCATION

Brown University | M.S. Cybersecurity (CS Track)

May 2028

Santa Clara University | B.S. Computer Science (Cybersecurity Emphasis)

June 2026

- Relevant Coursework: Operating Systems; Computer Security; Algorithms, Object-Oriented Programming, Data Structures (C++); Cryptography; Embedded Systems (ARM); Web Usability and Infrastructure (XML), Calculus I-IV; Discrete Math; Applied Cryptography

PUBLICATIONS & PRESENTATIONS

Aqlan, F., Kabadi, S., *Design Principles and Usability Analysis of Virtual Reality Manufacturing Digital Twins*: Innovation MDPI Journal, 2026 (In-progress)

Kabadi, S., Houshmand, S., *ML Framework for Password Policies*: Poster accepted to ACMW Computing Conference

Kabadi, S., Shaghaghi, N., *Hydration Automation: LoRa-based Smart Irrigation for Community Farms*: Poster presented at IEEE MIT URTC, IEEE GHTC

EXPERIENCE

NSF Research Assistant: VR Development and Data Analysis

May 2025 - Present

University of Louisville REU (NSF-funded) | Louisville, KY

- Developed a virtual reality environment in Unity using Photon Fusion to simulate a digital twin system, and collected eye-tracking data and user-interaction events to study task performance within the environment
- Implemented TLS-encrypted server connections, JWT authentication, and role-based access control to protect biometric information, and maintain the integrity of data access pathways. Engineered Python-based data analysis scripts (Pandas, Seaborn, Matplotlib) to process user-interaction logs
- Conducted a usability study and investigated the relationship between cognitive workload, VR familiarity, and system usability using usability perception (SUS) and cognitive load (SIM-TLX) scores.
- Proposed data-driven design improvements through a journal paper (2nd author– MDPI Journal) and a poster (IMECE 2025)

Student Researcher in Cybersecurity: Machine Learning and Password Security

December 2024 - March 2025

Santa Clara University | Santa Clara, CA

- Conducted a literature review and comparative analysis of existing password-strength frameworks (zxcvbn and LUDS), to study how deterministic rules and pattern-matching approaches overlook human cognitive habits and bias
- Utilized NLP techniques and trained a BERT model on large real-world password datasets to examine the linguistic and contextual patterns present in manually created credentials
- Fine-tuned a pre-trained BERT (bert-base-cased) model on 470K labeled passwords to classify strength levels (weak, medium, strong), achieving 87% accuracy and 0.81 F1 score on test data. Implemented model training, tokenization, and evaluation using PyTorch and scikit-learn to optimize learning rates, batching, and GPU performance
- Performed correlation analysis and K-Means clustering to evaluate the alignment between my BERT model's predictions and existing security approaches, identifying where modern meters (e.g., zxcvbn) differ structurally.
- Utilized these findings to argue that current password policy frameworks insufficiently account for behavioral and linguistic patterns in user-generated passwords, and suggested a more thorough framework.

UI/UX Researcher: Digital Consent and Privacy Policies

January 2025 - March 2025

Santa Clara University | Santa Clara, CA

- Investigated how interface design and dark patterns in Terms of Service (ToS) agreements infringes user consent
- Designed a prototype built on three core features: a plain-language quiz that reframes legal clauses as direct personal questions, a color-coded risk scale that visually flags concerning clauses within the contract itself, and alternative service suggestions for users who decide a given ToS is unacceptable
- Conducted usability research through user interviews to study how existing ToS agreements fail to produce genuine consent, and used the findings to argue that user-centric interface designs are crucial for digital autonomy.

Student Researcher in Internet of Things

May 2023 - Present

Ethical, Pragmatic, and Intelligent Computing Laboratory | Santa Clara, CA

- Developed a RESTful API using Node.js and Express to log IoT sensor data (moisture, humidity, temperature) into a SQL database
- Created a web-based dashboard using React and JavaScript to visualize real-time soil moisture data from LoRa sensors
- Collaborated with farmers at Veggielution; conducted meetings to refine the dashboard's usability; ensured the system aligned with the farm's needs

PROJECTS

Task Tracker | SQL, PHP, SQLite, JavaScript, HTML/CSS

- Created a web application with user authentication, task management, and progress tracking abilities using HTML, CSS, JavaScript, and SQL
- Integrated security measures and data integrity with hashed passwords, session management, and a SQLite Database

TECHNICAL SKILLS

Programming Languages: C/C++/C#, Python, Java/JavaScript, HTML/CSS, SQL, PHP

Software Tools: Tableau, Power BI, Jira, Confluence, Ubuntu, Docker, Unity, Unreal Engine, Postman, Wireshark, John the Ripper, Hashcat

Frameworks: React, Node.js, Express.js, Django, Scikit-learn, Transformers Library, PyTorch, Pandas, Matplotlib, Seaborn